

# Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

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## Abstract

Between 70–96 % of maritime casualties involve human error [1], and heavy weather makes coordination harder by degrading both perception and bridge-to-bridge radio. We present a tick-based agent-based model of Baltic and North Sea traffic. Vessels follow depth- and lane-constrained routes built from open bathymetry and Automatic Identification System (AIS) density data. Encounters are resolved with a Closest Point of Approach (CPA) check and a COLREGs-style give-way rule (International Regulations for Preventing Collisions at Sea). The main modelling choice is that every collision warning succeeds only with probability equal to the product of a local weather factor and a crew-fatigue factor. Collision damage uses DTU SIMCOL [2]; route-network collision frequency is checked with IWRAP Mk II [3]; rescue follows IAMSAR practice [4] with MASSIM/Koopman random search [5, 6] and Ashrafi seasonal degradation [7]. We compare a classical baseline, a weather-aware baseline, and a fatigue-aware Proposed System across four scenarios (30 seeds each; 360 runs) with Mann–Whitney U tests, Holm–Bonferroni correction, and Cliff’s  $\delta$ . Pre-registered fatal-rate and survival thresholds (H1–H3) were not confirmed after correction. Preparedness  $P_{\text{prep}}$  rose with large confirmed effects in every scenario ( $|\delta| \geq 0.90$ ). Mean Storm Corridor collisions fell by about 28 % versus Baseline A ( $\delta=0.32$ , not significant after Holm).

**Index Terms:** maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

## 1. Introduction

Maritime transport moves most of world trade, but safety remains limited by human factors. The International Maritime Organization (IMO)<sup>1</sup> attributes 70–96 % of incidents to human causes [1]. The European Maritime Safety Agency (EMSA) [8] reports weather as a frequent compounding factor in serious casualties, especially in congested coastal waters. Hull and machinery claims remain large [9].

The International Regulations for Preventing Collisions at Sea (COLREGs) [10]<sup>2</sup> assume that crews see the encounter and can agree who gives way. That assumption breaks down under two effects that many maritime ABMs treat only loosely [11, 12, 13]:

- **Weather and radio.** Sea clutter and storms reduce the

chance that a marine VHF<sup>3</sup> warning is heard and acted on. Many models assume that once risk is detected, the manoeuvre always happens.

- **Crew fatigue.** Fatigue builds over a watch and follows a circadian pattern [14]. Prior work often keeps fatigue in a side model, without a direct link to whether a warning is executed.

This paper couples those effects in one product: warning success depends on both local weather and crew fatigue. The model runs on real Baltic–North Sea geography (Section 3). Routes are synthesised from bathymetry and Automatic Identification System (AIS)<sup>4</sup> traffic density (Section 3.3). Avoidance uses Closest Point of Approach / Time to CPA (CPA/TCPA)<sup>5</sup> and a simple give-way rule (Section 4). Damage uses the DTU ship-collision model SIMCOL [2] (Section 5); frequency is checked with IALA IWRAP [3] (Section 7); survivors enter a search-and-rescue (SAR)<sup>6</sup> chain (Section 6). The engine is Rust on krABMaga [15]. Figure 1 summarises the causal chain.

## 2. Objectives and Hypotheses

We test whether linking fatigue management to warning reliability reduces fatal outcomes more than classical, weather-unaware practice. We implement three configurations:

**Baseline A (Classical)** COLREGs give-way, no weather-aware slowdown, unmanaged fatigue. Storm radio loss still applies inside the storm zone — it is environmental, not a privilege of other methods.

**Baseline B (Shore Rest Alerts)** Weather-aware speed reduction and slower fatigue build-up; no further mitigation.

**Proposed System** Smart watch scheduling (lowest fatigue build-rate) on top of Baseline B. Six KPIs are defined in Table 3.

We use them for the following hypotheses:

- H1.** Proposed reduces the fatal-event rate (Table 3, KPI 1) by  $\geq 20\%$  versus Baseline A ( $p < 0.05$ , Mann–Whitney U [16], Cliff’s  $|\delta| \geq 0.2$  [17]).
- H2.** Proposed improves post-incident survival (KPI 4) by  $\geq 15\%$  versus Baseline A.
- H3.** Under Storm Corridor, Proposed beats Baseline B on the fatal-event rate ( $p < 0.05$ ).

<sup>3</sup>[https://en.wikipedia.org/wiki/Marine\\_VHF\\_radio](https://en.wikipedia.org/wiki/Marine_VHF_radio)

<sup>4</sup>[https://en.wikipedia.org/wiki/Automatic\\_identification\\_system](https://en.wikipedia.org/wiki/Automatic_identification_system)

<sup>5</sup>Standard radar/AIS encounter metrics: distance and time to the predicted closest point of approach. Overview: [https://en.wikipedia.org/wiki/Collision\\_avoidance](https://en.wikipedia.org/wiki/Collision_avoidance)

<sup>6</sup>[https://en.wikipedia.org/wiki/Search\\_and\\_rescue](https://en.wikipedia.org/wiki/Search_and_rescue)

<sup>1</sup>[https://en.wikipedia.org/wiki/International\\_Maritime\\_Organization](https://en.wikipedia.org/wiki/International_Maritime_Organization)

<sup>2</sup>[https://en.wikipedia.org/wiki/International\\_Regulations\\_for\\_Preventing\\_Collisions\\_at\\_Sea](https://en.wikipedia.org/wiki/International_Regulations_for_Preventing_Collisions_at_Sea)

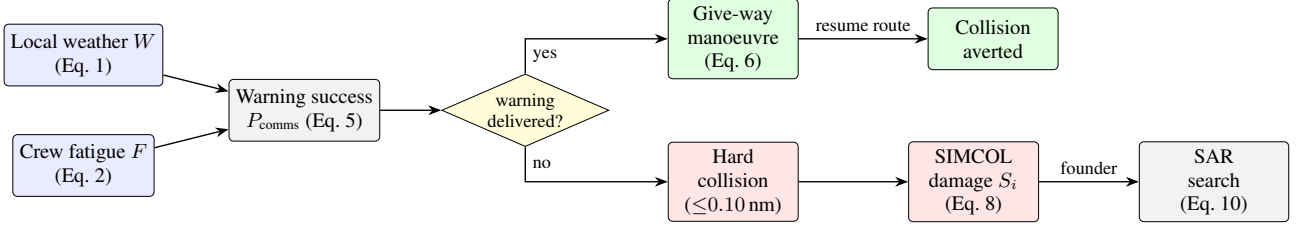


Figure 1: *Causal pipeline.* Weather  $W$  and fatigue  $F$  set warning success  $P_{\text{comms}}$  (Eq. 5). A delivered warning yields a give-way manoeuvre; a missed warning leaves the encounter to geometry. Hard collisions are scored with SIMCOL; a foundering ship enters SAR.

### 3. Simulation Model

This section defines the world, agents, routes, and weather field used by all methods.

#### 3.1. Domain and Time

The domain is the Baltic and North Sea: latitude  $\phi \in [50.5^\circ, 66.0^\circ]$  N and longitude  $\lambda \in [-5.0^\circ, 31.0^\circ]$  E. An equirectangular<sup>7</sup> map with a mid-latitude correction sets one field unit to one nautical mile<sup>8</sup> at  $\phi_0 = 58.25^\circ$ :

$$x = (\lambda - \lambda_{\min}) k_{\text{lon}}, \quad y = (\phi - \phi_{\min}) k_{\text{lat}},$$

with  $k_{\text{lat}} = 60 \text{ nm}^\circ$  and  $k_{\text{lon}} = 60 \cos \phi_0$ . East–west distortion at the box edges is about 6–10 % and is left uncorrected; distances inside the engine are Euclidean nautical miles. Coastline polygons from Natural Earth<sup>9</sup> sit in an R-tree<sup>10</sup> for land and grounding tests. Bathymetry is used only offline for route planning (Section 3.3).

Time steps are  $\Delta t = 15 \text{ min}$  (0.25 h). The default run length is  $T = 2880$  ticks (30 days). A wall-clock variable starts at 06:00 UTC and drives the circadian fatigue term (Section 4). Each run uses one seeded `SmallRng`, so the same seed yields identical KPI logs.

#### 3.2. Vessels and Voyage Logic

Agents are ships; after an incident, rescue assets appear (Section 6). Ship  $i$  has position  $\mathbf{p}_i$ , waypoint route  $R_i$ , travel direction  $d_i \in \{+1, -1\}$ , heading  $\psi_i$ , cruise speed  $v_i$ , state  $\sigma_i \in \{\text{ACTIVE}, \text{DOCKED}\}$ , crew  $n_i = 10$ , and fatigue  $F_i \in [0, 1]$ .

Routes are synthesised once (Section 3.3) and stored as `data/ais_paths.json`. Ports are polygons in `data/ports.json`. Entering a port docks the ship. Cruise speed is drawn at spawn ( $u \sim \mathcal{U}(0, 1)$ ):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / ro-ro / ro-pax} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases}$$

**Waypoint pursuit.** The active target is an avoidance waypoint if one is queued, otherwise the current route waypoint. A waypoint is reached within 0.5 nm. The step length is  $s = \min(v_i \Delta t, \rho)$ . Moves use up to five geometrically halved

lengths so thin land is not tunnelled; a post-move check reverts grounding to the last valid position. At a port or route end the ship dwells  $D \sim \mathcal{U}\{8, 48\}$  ticks, then reverses. Fleet size  $N$  is held fixed: when a ship is removed, a new one is spawned on a random synthesised track.

**Same-destination following.** Co-directional ships with end-points within 5 nm share a destination. A trailer within 3 nm of a leader matches the leader’s speed; within 0.5 nm it holds station. This avoids side-by-side overlap that the give-way rule does not separate.

#### 3.3. Route Generation

Traffic geometry is produced offline so experiments stay reproducible. Depth comes from the EMODnet DTM [18]. Lane attractiveness comes from EMODnet vessel-density maps [19, 12]. Both are fetched once via the OGC Web Coverage Service (WCS)<sup>11</sup> and stored as grids. Ports are a gazetteer of 42 harbours snapped to navigable water.

Each class  $c$  has draft  $T_c$  and under-keel clearance (UKC <sub>$c$</sub> )<sup>12</sup> [20]. A cell of depth  $h$  is navigable iff  $h \geq T_c + \text{UKC}_c + m_s$  with buffer  $m_s = 2 \text{ m}$  (Table 1). Deep-draft very large crude carriers (VLCCs)<sup>13</sup> are excluded (Danish Straits).

Table 1: *Minimum transit depth by vessel class (draft + under-keel clearance).*

| Class     | Draft $T_c$ (m) | UKC <sub><math>c</math></sub> (m) | Required $h$ (m) |
|-----------|-----------------|-----------------------------------|------------------|
| Passenger | 6.5             | 1.5                               | 8                |
| Cargo     | 11.0            | 2.0                               | 13               |
| Tanker    | 14.0            | 3.0                               | 17               |

Between sampled port pairs, a weighted A\* planner [21] runs on a 1 nm grid. Cell cost uses

$$\mu = 1 + w_h s_h + w_d s_d + w_\ell (1 - a_\ell),$$

penalising shallow margins ( $s_h$ ), proximity to shore/shoal ( $s_d$ ), and off-lane cells ( $a_\ell$ ). Default weights are  $(w_h, w_d, w_\ell) = (3, 4, 2.5)$ . Paths are simplified under a depth constraint and checked against raw bathymetry every 0.15 nm. Seeded cost jitter keeps outputs deterministic per seed. The default network has 80 routes (35 cargo, 25 passenger, 20 tanker).

<sup>11</sup>[https://en.wikipedia.org/wiki/Web\\_Coverage\\_Service](https://en.wikipedia.org/wiki/Web_Coverage_Service)

<sup>12</sup>[https://en.wikipedia.org/wiki/Under\\_keel\\_clearance](https://en.wikipedia.org/wiki/Under_keel_clearance)

<sup>13</sup>[https://en.wikipedia.org/wiki/Very\\_large\\_crude\\_carrier](https://en.wikipedia.org/wiki/Very_large_crude_carrier)

<sup>7</sup>[https://en.wikipedia.org/wiki/Equirectangular\\_projection](https://en.wikipedia.org/wiki/Equirectangular_projection)

<sup>8</sup>[https://en.wikipedia.org/wiki/Nautical\\_mile](https://en.wikipedia.org/wiki/Nautical_mile)

<sup>9</sup>[https://en.wikipedia.org/wiki/Natural\\_Earth](https://en.wikipedia.org/wiki/Natural_Earth)

<sup>10</sup><https://en.wikipedia.org/wiki/R-tree>

### 3.4. Weather Field

A  $50 \times 50$  grid holds sea state  $s$ , visibility  $v$ , and wind  $u$  in  $[0, 1]$ . They fuse into hazard  $W_{ij} \in [0, 1]$ :

$$W = \text{clip} \left( \frac{w_s s + w_v (1 - v) + w_u u}{w_s + w_v + w_u}, 0, 1 \right), \quad (1)$$

with  $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$ . Each ship samples the nearest cell. The field advances each tick through a calm/storm regime chain, Lagrangian storm cells, AR(1)<sup>14</sup> background fields, overlays, coupling, and smoothing (parameters in Appendix A).

Independently, Storm Corridor places a moving storm of radius  $r_{\text{storm}}$  that drifts at 0.30 nm/tick along a fixed track (English Channel → North Sea → Skagerrak → Kattegat → Baltic → Gdańsk). Inside the zone, communication and (for non-A methods) speed are degraded.

## 4. Crew Fatigue and Collision Avoidance

Fatigue and avoidance form the mechanism behind H1–H3: warnings fail when weather is bad and crews are tired.

### 4.1. Fatigue Dynamics

Fatigue  $F_i \in [0, 1]$  updates every tick. The circadian drive peaks near 03:00:

$$c(t_{\text{utc}}) = \frac{1}{2} \left( 1 - \cos \left( \frac{2\pi(t_{\text{utc}} - 3)}{24} \right) \right) \in [0, 1].$$

While ACTIVE, with local hazard  $W$ ,

$$\Delta F = (\beta_{\text{base}} + \beta_W W + \beta_c c(t_{\text{utc}})) m(\text{method}), \quad (2)$$

with  $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015, 0.0040, 0.0008)$  and

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A} \\ 0.82 & \text{Baseline B} \\ 0.65 & \text{Proposed,} \end{cases} \quad (3)$$

then  $F_i \leftarrow \min(1, F_i + \Delta F)$ . While DOKED, recovery is 0.025 per tick.

Above threshold  $F^* = 0.40$ , fatigue reduces warning reliability with maximum penalty  $\kappa = 0.65$ :

$$g(F_i) = \begin{cases} 1, & F_i \leq F^* \\ 1 - \kappa \frac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35, 1]. \quad (4)$$

### 4.2. CPA Detection and Give-Way

Each ACTIVE pair is screened with linear CPA. With relative position  $\Delta \mathbf{p}$  and relative velocity  $\Delta \mathbf{v}$ ,

$$t^* = - \frac{\Delta \mathbf{p} \cdot \Delta \mathbf{v}}{\|\Delta \mathbf{v}\|^2}, \quad d_{\text{CPA}} = \|\Delta \mathbf{p} + t^* \Delta \mathbf{v}\|,$$

where  $t^*$  is in ticks. A manoeuvre is considered only if  $\|\Delta \mathbf{p}\| \leq 20$  nm,  $d_{\text{CPA}} \leq 5$  nm,  $0 < t^* \leq 12$  ticks, and the give-way ship is not in cooldown.

<sup>14</sup>[https://en.wikipedia.org/wiki/Autoregressive\\_model](https://en.wikipedia.org/wiki/Autoregressive_model)

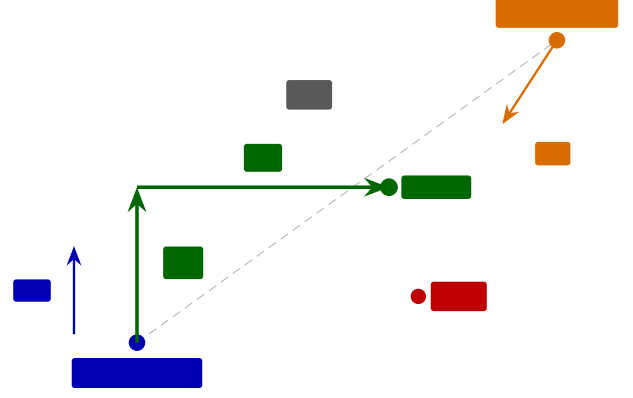


Figure 2: Give-way geometry. Ship a steers to a temporary waypoint  $\mathbf{w}_{\text{avoid}}$  placed  $L_f$  ahead and  $L_o$  to starboard (Eq. 6). The manoeuvre runs only if the warning is delivered (Eq. 5).

The lower-id ship gives way (turn to starboard); the other stands on. This is a reproducible surrogate for COLREGs stand-on/give-way assignment [10, 11]. The manoeuvre runs only if the warning is delivered:

$$P_{\text{comms}} = \underbrace{\min(q(\mathbf{p}_a), q(\mathbf{p}_b))}_{\text{weather / storm}} \cdot \underbrace{\min(g(F_a), g(F_b))}_{\text{fatigue (Eq. 4)}}, \quad (5)$$

where  $q(\mathbf{p})$  is nominal link reliability, reduced to the storm-comms rate (0.08 inside Storm Corridor) when  $\mathbf{p}$  is in the storm zone. A uniform draw below  $P_{\text{comms}}$  executes the manoeuvre; otherwise geometry alone decides the encounter. Equation 5 is the central coupling of the model.

A give-way ship receives one temporary waypoint  $L_f = 3$  nm ahead and  $L_o = 5$  nm to starboard of heading  $\psi$  (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f (\sin \psi, \cos \psi) + L_o (\cos \psi, -\sin \psi). \quad (6)$$

A short message log records the warning exchange. A 3-tick cooldown prevents thrashing.

### 4.3. Optional Force-Field Track

As an optional, default-off switch, the discrete waypoint of Eq. 6 can be replaced by a continuous repulsion field in the style of Xiao et al. [22]:

$$\mathbf{F}_{\text{rep}} = \sum_{k: d_k < d_o} \frac{k_{\text{rep}}}{d_k^p} \hat{\mathbf{n}}_k,$$

with exponent  $p = 2$  by default. Source effect distances are micro-scale relative to a 15-minute basin tick, so magnitudes are rescaled while keeping the reported head-on to overtaking ratio. All definitive experiments keep this track off.

## 5. Collision Consequence (SIMCOL)

A **hard collision** is logged when two ACTIVE ships close within 0.10 nm ( $\approx 185$  m), once per encounter (rising edge). Contacts within 1 nm of a port are ignored as harbour manoeuvring under vessel traffic services (VTS)<sup>15</sup>. Damage uses DTU SIMCOL (ship collision damage / survivability) [2].

<sup>15</sup>[https://en.wikipedia.org/wiki/Vessel\\_traffic\\_service](https://en.wikipedia.org/wiki/Vessel_traffic_service)

Absorbed energy follows Pedersen and Zhang [23, 2] on the normal closing component:

$$E_{\text{abs}} = \frac{1}{2} \frac{M'_a M'_b}{M'_a + M'_b} V_n^2, \quad V_n = |(\mathbf{v}_b - \mathbf{v}_a) \cdot \hat{\mathbf{n}}|, \quad (7)$$

with added-mass factors  $m_{\text{sway}} = 0.85$  and  $m_{\text{surge}} = 0.05$  [2]. Minorsky's correlation [24] links energy to damaged volume. Relative penetration  $t/B$  drives a survival factor shaped like SOLAS Chapter II-1 probabilistic damage stability [25]<sup>16</sup>:

$$S_i = \begin{cases} 1, & t/B \leq r_0 \\ \left( \frac{r_1 - t/B}{r_1 - r_0} \right)^{0.25}, & r_0 < t/B < r_1 \\ 0, & t/B \geq r_1, \end{cases} \quad (8)$$

with  $r_0 = 0.10$ ,  $r_1 = 0.30$ . This is an accepted surrogate: the engine does not carry per-ship residual-stability curves. Expected overboard count is

$$\mathbb{E}[\text{overboard}] = n_b (1 - S_i), \quad (9)$$

drawn as independent Bernoulli trials. If  $S_i < S^\dagger = 0.50$ , the struck ship founders and the whole crew enters the liferaft (EVAC). If it stays afloat, only the overboard draw enters the man-overboard chain (Section 6.3).

Encounter type is diagnostic, from the folded heading angle  $\theta$ :

$$\text{type} = \begin{cases} \text{side-to-side (overtaking)}, & \theta < 45^\circ \\ \text{front-to-side (crossing)}, & 45^\circ \leq \theta < 135^\circ \\ \text{head-on}, & \theta \geq 135^\circ. \end{cases}$$

Near-parallel hits release little energy in Eq. 7; head-on and crossing hits drive deeper penetration.

## 6. Search and Rescue

Foundering ( $S_i < S^\dagger$ ) starts the liferaft chain, following International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual practice [4]<sup>17</sup>. A rescue asset leaves the nearest port after a mobilisation delay of 2 ticks by default: helicopter (80 kn) beyond 40 nm offshore, otherwise patrol (25 kn). A second chain handles persons overboard from afloat ships (Section 6.3). Figure 3 shows the vessel state machine. Grounding only reverts position; it does not force evacuation.

### 6.1. Drift and Detection (MASSIM)

Detection follows the Koopman random-search model used in the MASSIM maritime-search simulation [5, 6]. With sweep width  $w$ , search speed  $v_s$ , and area  $A$ ,

$$\text{CDP} = 1 - \exp\left(-\frac{n w v_s \tau}{A}\right) = 1 - e^{-C}. \quad (10)$$

Search area grows as  $A(\tau) = \pi(r_0 + u_d \tau)^2$  with initial radius  $r_0$  and drift  $u_d$ . Each on-scene tick draws Bernoulli detection at coverage  $1 - e^{-dC}$ .

<sup>16</sup>[https://en.wikipedia.org/wiki/International\\_Convention\\_for\\_the\\_Safety\\_of\\_Life\\_at\\_Sea](https://en.wikipedia.org/wiki/International_Convention_for_the_Safety_of_Life_at_Sea)

<sup>17</sup>[https://en.wikipedia.org/wiki/International\\_Aeronautical\\_and\\_Maritime\\_Search\\_and\\_Rescue\\_Manual](https://en.wikipedia.org/wiki/International_Aeronautical_and_Maritime_Search_and_Rescue_Manual)

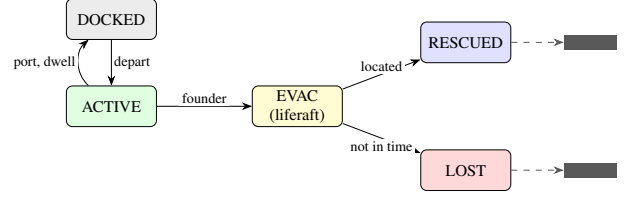


Figure 3: *Vessel state machine. Foundering collision → EVAC; SAR yields RESCUED (Eq. 10) or LOST. Terminal ships respawn to keep fleet size. KPI fatalities count only unrecovered persons-in-water (Section 6.3).*

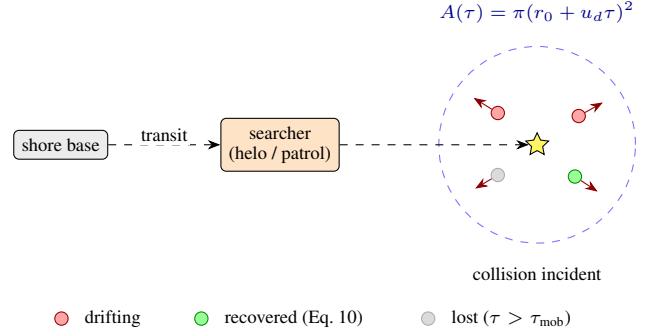


Figure 4: *Man-overboard search. Casualties (Eq. 9) drift separately; one searcher applies Eq. 10 per person. Unrecovered people within  $\tau_{\text{mob}}$  are the fatal KPI events.*

### 6.2. Seasonal Degradation (Ashrafi)

Asset performance follows Ashrafi et al. [7]. Wind-chill uses

$$\text{wci} = 13.12 + 0.6215 T - 11.37 U^{0.16} + 0.3965 T U^{0.16}.$$

Months map to severity groups A–D (May–August mildest; Dec–Feb harshest). Relative transit/search speeds are approximately 1.00/0.95/0.77/0.60 (helicopter) and 1.00/0.80/0.40/0.21 (patrol); boarding times lengthen with severity [7]. Default simulation month is July (group A).

### 6.3. Man-Overboard Search

Overboard casualties (Eq. 9) form one incident of persons-in-water, each with its own drift bearing (Fig. 4), following Karatas et al. [5]. One searcher homes on the cluster centroid and applies Eq. 10 independently per person. Anyone not recovered within  $\tau_{\text{mob}} = 12$  ticks ( $\approx 3$  h) is lost. **This is the only path that increments the fatal KPI.** Liferaft LOST is logged but does not add to fatal\_per\_1k\_hrs. Liferaft endurance is 96 ticks (24 h).

## 7. External Collision-Frequency Check (IWRAP)

Internal KPIs compare methods. To place traffic intensity on a known scale, the committed route network is analysed with IALA's IWRAP Mk II waterway risk model [3, 26]<sup>18</sup>:

$$N_c = N_a P_c.$$

<sup>18</sup>[https://en.wikipedia.org/wiki/International\\_Association\\_of\\_Marine\\_Aids\\_to\\_Navigation\\_and\\_Lighthouse\\_Authorities](https://en.wikipedia.org/wiki/International_Association_of_Marine_Aids_to_Navigation_and_Lighthouse_Authorities)

Head-on, overtaking, and crossing candidate counts follow the usual IWRAP forms (flows  $Q$ , beams  $B$ , speeds  $V$ , Pedersen diameter for crossings [3, 27]). Causation probabilities are the IWRAP defaults:  $P_c^{\text{head-on}} = 0.50 \times 10^{-4}$ ,  $P_c^{\text{overtaking}} = 1.10 \times 10^{-4}$ ,  $P_c^{\text{crossing}} = 1.30 \times 10^{-4}$ .

Each batch row stamps  $N_c$  from the static synthesised routes, the scenario fleet size, and a method speed scale (weather-aware methods slow traffic). We do not export per-run AIS tracks. For the present network,  $N_c$  is on the order of 0.01 collisions/yr, inside the usual  $<1$ /yr/fairway acceptance band.

## 8. Experimental Design

### 8.1. Methods and Scenarios

The three methods of Section 2 differ as follows:

- **Baseline A:** no weather-aware slowdown (storm speed factor = 1); unmanaged fatigue ( $m = 1.00$ ).
- **Baseline B:** hazard and storm slowdowns;  $m = 0.82$ .
- **Proposed:** same slowdowns as B;  $m = 0.65$ .

SIMCOL parameters are identical across methods. Storm-zone radio loss applies to all methods. Table 2 lists the four scenarios. Each condition uses 30 seeds and 2880 ticks ( $4 \times 3 \times 30 = 360$  runs).

### 8.2. Key Performance Indicators

Table 3 defines the six hypothesis KPIs. Ship-hours add 0.25 h per ACTIVE ship per tick. A *fatality* in the KPIs is an unrecovered person-in-water (Section 6.3).

Crew preparedness uses method awareness  $\alpha_M \in \{2.0 \text{ (A)}, 1.3 \text{ (B)}, 1.0 \text{ (Proposed)}\}$ :

$$P_{\text{prep}} = \max(0, 1 - \alpha_M W) \cdot \max(0, 1 - 0.70 F).$$

Diagnostic counters (collision-type mix, MOB tallies) are logged but not entered into the Holm battery.

### 8.3. Statistical Analysis

For each (KPI  $\times$  scenario  $\times$  method pair) we use a two-tailed Mann–Whitney U test [16] with Cliff’s  $\delta$  [17]. Multiplicity uses Holm’s sequentially rejective Bonferroni procedure [28] over the full battery. A result is *significant* if  $p_{\text{corr}} < 0.05$  and *confirmed* if also  $|\delta| \geq 0.2$ . Pairs are (A vs Proposed), (B vs Proposed), and (A vs B). Seeds are matched across conditions.

## 9. Implementation

The engine is Rust on krABMaga [15]. An Axum HTTP API launches factorial batches (Rayon) and streams progress with Server-Sent Events (SSE)<sup>19</sup>. Each finished run writes a JSONL tick log (every ten ticks), a manifest, and a SQLite<sup>20</sup> results row. A React workbench (deck.gl / MapLibre) replays completed runs.

### 9.1. Tick Ordering

Within a tick, vessel agents step first (krABMaga order 200), then a post-tick agent (order 1000) runs the shared physics pipeline shown in Fig. 5: weather advance; grounding revert;

<sup>19</sup>[https://en.wikipedia.org/wiki/Server-sent\\_events](https://en.wikipedia.org/wiki/Server-sent_events)

<sup>20</sup><https://en.wikipedia.org/wiki/SQLite>

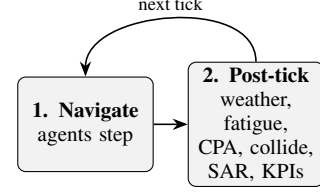


Figure 5: *Macro-tick order. Agents navigate, then the post-tick pipeline updates weather, human factors, encounters, and outcomes.*

hazard and storm speed damping; fatigue; CPA avoidance and comms draws; following; rising-edge collisions and SIMCOL; SAR/MOB; fleet top-up; KPI accumulate; optional JSONL frame. Navigation therefore uses weather and avoidance state from the previous tick’s post-tick pass.

### 9.2. Speed Reduction

Baseline B and Proposed interpolate the just-moved position toward the last valid position by

$$f_W = \max(0.30, 1 - 0.45 W).$$

Baseline A ignores this factor. Inside the storm zone an extra factor applies: 1.0 for Baseline A, 0.45 in Storm Corridor for the other methods (default non-corridor storm factor 0.70 when a storm is enabled).

## 10. Results

Confirmation requires  $p_{\text{corr}} < 0.05$  and  $|\delta| \geq 0.2$ . Figure 6 shows per-run KPI distributions; Figure 7 summarises effect sizes for the pre-registered contrasts.

### 10.1. Calibration and Storm Contrast

Under Baseline A, Calm Passage mean collisions were 0.60 per 1 000 ship-hours (inside the 0.5–1.5 band). Storm Corridor raised that rate to 1.48 ( $2.48 \times$  calm), restoring a clear weather contrast after the depth+lane route rebase.

### 10.2. Hypotheses H1–H3

Table 4 summarises the pre-registered contrasts. **H1** and **H2** are not confirmed: fatal rates and survival ratios sit near floor/ceiling in Calm Passage, Storm Corridor, and Blind Shore, so tests have little power. Deep Water Rescue has a higher fatal base rate; Proposed reduces mean fatals from 0.038 (Baseline A) to 0.023 ( $\approx 39\%$ ), but survival is almost unchanged and neither contrast survives Holm ( $p_{\text{corr}} = 1$ ,  $|\delta| \leq 0.02$ ). **H3** is non-significant (means 0.0022 vs 0.0024).

### 10.3. Mechanism and Collisions

Mean  $P_{\text{prep}}$  is higher for Proposed than for either baseline in every scenario, with confirmed effects ( $|\delta| \in [0.90, 1.00]$ ). Mean Storm Corridor collisions fall from 1.48 to 1.07 versus Baseline A ( $\approx 28\%$ ,  $\delta = 0.32$ ), but that contrast is not significant after family-wise Holm correction.

## 11. Discussion

The core claim is Eq. 5: a warning succeeds only when weather and both crews allow it. Methods differ by fatigue build-rate

Table 2: *Scenario summary (30 seeds, 2880 ticks/run).*

| Scenario          | Key conditions                                                                                                      | Ships | Hyp.   |
|-------------------|---------------------------------------------------------------------------------------------------------------------|-------|--------|
| Calm Passage      | Calm weather; calibration reference (target band $\approx 0.5$ – $1.5$ collisions / 1 k ship-hrs under Baseline A). | 20    | Calib. |
| Storm Corridor    | Moving storm; storm-comms rate 0.08; non-A speed factor 0.45.                                                       | 25    | H1, H3 |
| Blind Shore       | Fleet-wide comms 0.55; tighter avoidance berth ( $L_o = 3$ nm).                                                     | 25    | H1     |
| Deep Water Rescue | Sparse fleet; SAR mobilisation 4 ticks; helo/patrol 55/16 kn.                                                       | 15    | H2     |

Table 3: *KPI definitions and hypothesis links.*

| # | KPI                  | Definition                                     | Hyp.   |
|---|----------------------|------------------------------------------------|--------|
| 1 | fatal_per_1k_hrs     | Fatalities $\div$ ship-hrs $\times 10^3$       | H1, H3 |
| 2 | collision_per_1k_hrs | Collisions $\div$ ship-hrs $\times 10^3$       | Calib. |
| 3 | evac_activation_rate | Evac activations $\div$ ship-hrs $\times 10^3$ | —      |
| 4 | survival_ratio       | 1 – fatalities / crew exposed in collisions    | H2     |
| 5 | avg_tta_hours        | Mean rescue time-to-arrival                    | H2     |
| 6 | mean_p_prep          | Time-averaged crew preparedness                | mech.  |

Table 4: *Pre-registered hypothesis outcomes (definitive 30-seed batch).*

| Hyp. | Contrast                          | Verdict       | Cliff’s $\delta$      |
|------|-----------------------------------|---------------|-----------------------|
| H1   | Fatal rate, Proposed vs A         | Not confirmed | $ \delta  \leq 0.07$  |
| H2   | Survival ratio, Proposed vs A     | Not confirmed | $ \delta  \leq 0.07$  |
| H3   | Fatals under Storm, Proposed vs B | Not confirmed | $\delta \approx 0.00$ |

$m$  (Eq. 3), weather/storm speed awareness, and preparedness weights  $\alpha_M$ . The batch confirms the preparedness lever and a directional storm-collision benefit, but not the pre-registered fatal/survival thresholds. Fatal events are rare outside Deep Water Rescue, so H1–H3 are under-powered even when point estimates move in the expected direction.

Accredited layers add external structure: SIMCOL for damage [2], IWRAP for network frequency [3], and MAS-SIM/Ashrafi for rescue [5, 7]. The survival factor  $S_i$  (Eq. 8) is a SOLAS-shaped surrogate, not a full residual-stability calculation.

Main limitations: synthesised routes trade some traffic realism for control; the lower-id give-way rule simplifies COLREGs; force-field avoidance is off by default; KPI fatalities count only MOB timeouts; IWRAP stamps use static routes rather than per-run tracks. Calibration under Baseline A Calm Passage (0.60 collisions per 1000 ship-hours) sits near published AIS near-miss bands [12, 13]. Longer runs or collision-rate primary endpoints would raise power on fatal contrasts.

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Definitive batch KPIs (30 seeds & 4 scenarios × 3 methods)

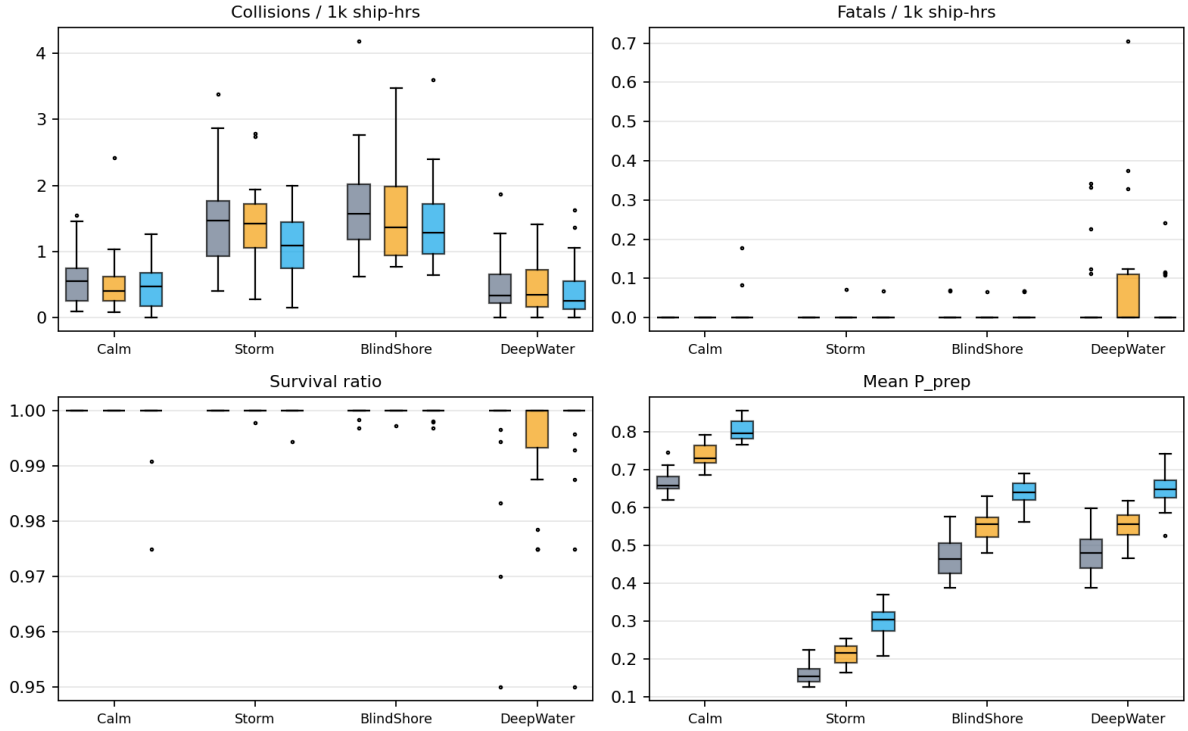


Figure 6: Per-run KPI distributions for the definitive batch (30 seeds). A = Baseline A, B = Baseline B, P = Proposed.

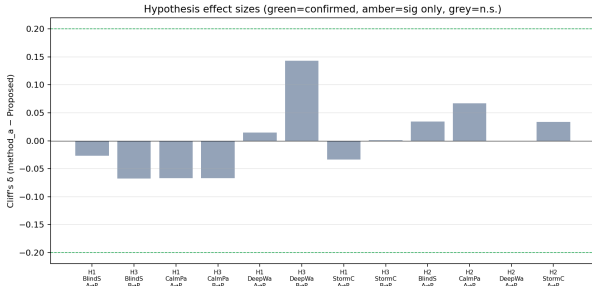


Figure 7: Cliff's  $\delta$  for H1–H3-relevant pairs. Green: confirmed; amber: significant only; grey: not significant after Holm.

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Table 5: Selected engine default constants.

| Constant                                  | Default                          | Role                                         |
|-------------------------------------------|----------------------------------|----------------------------------------------|
| $\Delta t$                                | 0.25 h                           | tick length (15 min)                         |
| $T$                                       | 2880                             | ticks per run (30 d)                         |
| weather grid                              | $50 \times 50$                   | hazard field                                 |
| $(w_s, w_v, w_u)$                         | (1.0, 0.5, 0.8)                  | hazard fusion weights                        |
| warn / CPA / TTA gate                     | 20 / 5 nm / 12 tk                | avoidance gate                               |
| $L_f, L_o$                                | 3 / 5 nm                         | give-way waypoint offsets                    |
| cooldown                                  | 3 tk                             | re-manoeuve lockout                          |
| $r_{\text{coll}}$                         | 0.05 nm                          | hit at $2r = 0.10$ nm                        |
| port collision exclusion                  | 1 nm                             | harbour water                                |
| follow vicinity / gap                     | 3 / 0.5 nm                       | speed-match / keep-distance                  |
| same-destination                          | 5 nm                             | endpoint match radius                        |
| $(m_{\text{sway}}, m_{\text{surge}})$     | 0.85, 0.05                       | SIMCOL added mass                            |
| Minorsky $a, b$                           | 47.2, 32.7                       | energy-volume (MJ, m <sup>3</sup> )          |
| $\ell / L_{pp}, \tau$                     | 0.12, 0.08 m                     | damage length / side steel                   |
| $r_0, r_1, S^\dagger$                     | 0.10, 0.30, 0.50                 | $S_i$ knee / loss / founder                  |
| $(\beta_{\text{base}}, \beta_W, \beta_c)$ | 0.0015, 0.0040, 0.0008           | fatigue rates                                |
| recovery                                  | 0.025/tk                         | dock fatigue recovery                        |
| $F^*, \kappa$                             | 0.40, 0.65                       | comms-degradation knee                       |
| storm-comms (corridor)                    | 0.08                             | Storm Corridor link rate                     |
| storm-comms (generic default)             | 0.30                             | when storm enabled out-side corridor presets |
| $n$ crew                                  | 10                               | per vessel                                   |
| dwell                                     | 8–48 tk                          | port stay                                    |
| helicopter / patrol                       | 80 / 25 kn                       | rescue asset speeds                          |
| $\Delta t_{\text{mob}}$                   | 2 tk                             | rescue mobilisation delay                    |
| helo / patrol cutoff                      | 40 nm                            | SAR asset-selection range                    |
| $w, r_0, u_d$                             | 5 nm, 2 nm, 0.2 nm/tk            | MASSIM sweep / datum / drift                 |
| survival window                           | 96 tk                            | liferaft endurance (24 h)                    |
| $\tau_{\text{mob}}$                       | 12 tk                            | MOB cold-water window (3 h)                  |
| MOB scatter                               | 0.5 nm                           | person-in-water spread                       |
| $P_c$ (IWRAP)                             | $0.5\text{--}1.3 \times 10^{-4}$ | causation probability                        |

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## A. Default Constants

Table 5 lists principal defaults (overridable in configuration).

Weather-field process (Mixed preset): regime flips  $p_{cs} = 0.03$ ,  $p_{sc} = 0.08$ ; Calm/Storm reversion targets (0.30, 0.80, 0.25) / (0.72, 0.30, 0.75); up to 8 storm cells; AR(1) coefficients  $(\rho_s, \rho_v, \rho_u) = (0.95, 0.90, 0.93)$ ,  $(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$ ; gradient cap 0.12. Calm and Stormy presets scale the flip rates.